Updated-Ensemble FP32 Calibration

New identity-disjoint cohort • same thresholds rules • no protected confirmation reuse

STRICT 0.25% FPR GATE: BOTH FAIL

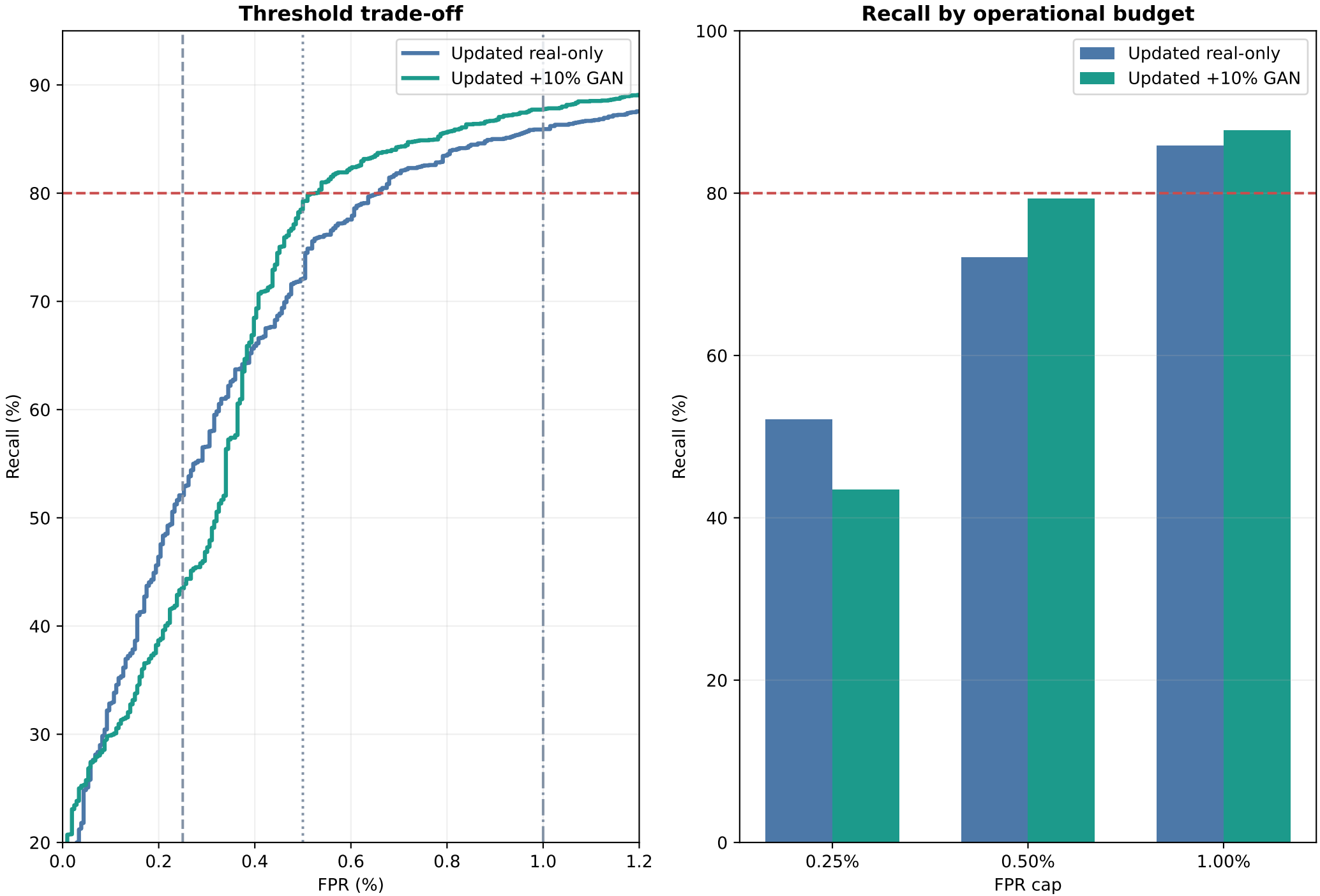
Condition	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Updated real-only	0.999411	96.30%	52.08%	67.60%	98.10%	92.66%	0.24%	47.92%
Updated +10% GAN	0.999849	95.52%	43.48%	59.76%	98.03%	92.71%	0.25%	56.52%

FP32 fixed score quantization, but not distribution shift.

Real-only reaches 52.08% recall and +10% GAN reaches 43.48% recall at the strict FPR budget. Their ROC-AUC values near 98% show ranking, but the low-FPR tail is not sufficiently separated to meet the registered browser-extension gate.

This calibration result supersedes the uncalibrated internal thresholds for deployment discussion; it does not alter the internal experiment its

FP32 Calibration Curves



Decision and Next Action

What the corrected calibration supports

FPR cap 0.25%

Updated real-only: recall 52.08%, F1 67.60%, projected precision at 1% prevalence 68.43%

Updated +10% GAN: recall 43.48%, F1 59.76%, projected precision at 1% prevalence 63.95%

FPR cap 0.50%

Updated real-only: recall 72.12%, F1 81.84%, projected precision at 1% prevalence 59.30%

Updated +10% GAN: recall 79.28%, F1 86.46%, projected precision at 1% prevalence 61.56%

FPR cap 1.00%

Updated real-only: recall 85.84%, F1 88.57%, projected precision at 1% prevalence 47.18%

Updated +10% GAN: recall 87.72%, F1 89.62%, projected precision at 1% prevalence 47.59%

VERDICT

Neither ensemble meets 80% recall at FPR <= 0.25%. Real-only is better at the strict cap; +10% GAN becom
FPR budget is relaxed to 0.5% or 1%. FP32 scoring corrected threshold resolution but did not solve low-FPR
shift.

Do not choose the 0.5% or 1% setting merely because it reaches higher recall. The operational FPR budget must be justified before viewing a future confirmation set. If the project retains 0.25%, the model representation/training data—not just calibration—must improve.